Smart Device Charging Hub Manager

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Module: Data structures and algorithms

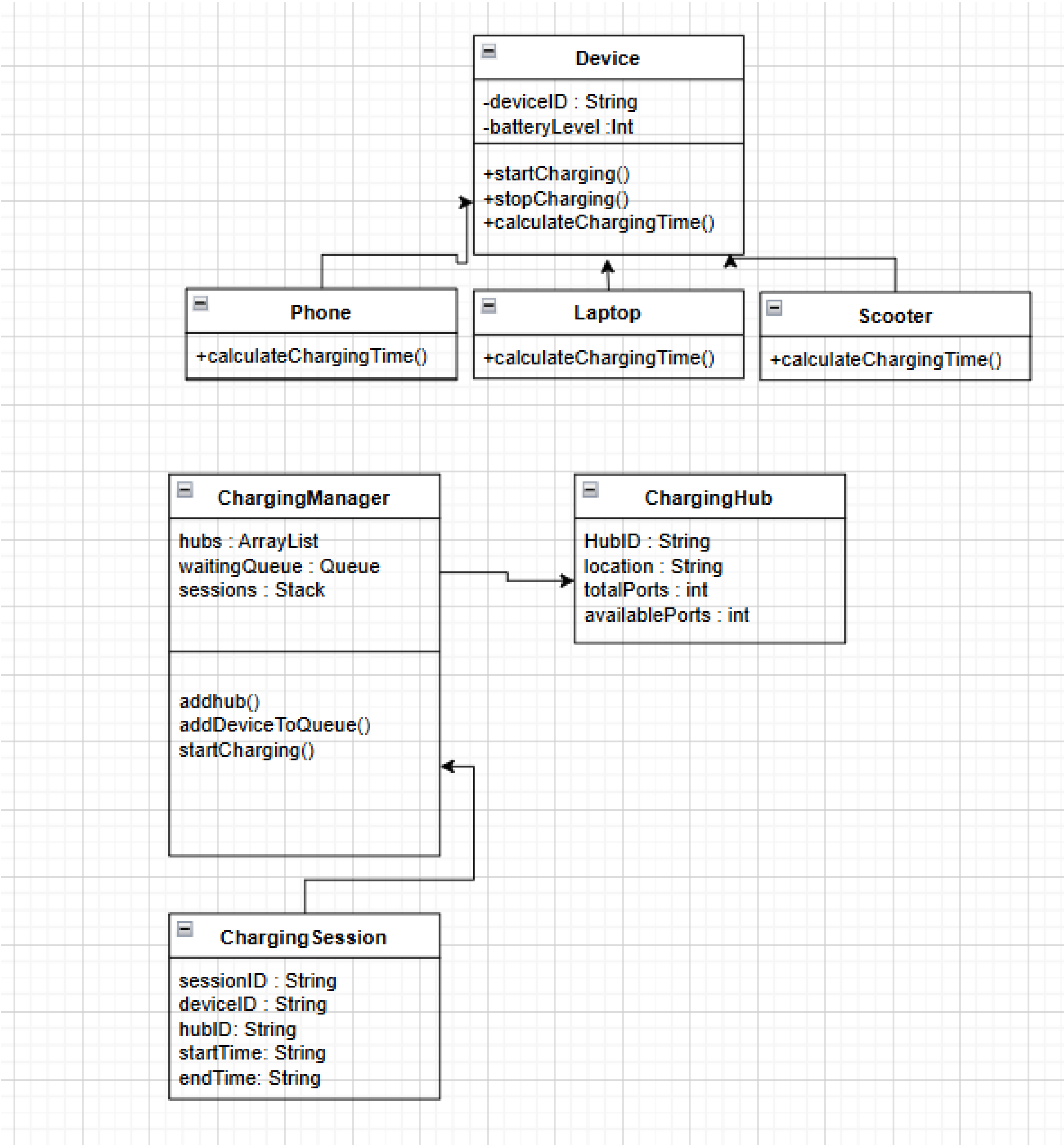
Chosen Smart Docklands Theme

The theme selected for this project is Smart Mobility and a smart design/layout from the Smart Docklands Pilot Call. Cities are now more dependent on mobile technology and electric mobility devices such as e-scooters and e-bikes. This now results in a need for public charging spaces/ charging ports in a given area.

Application Description

The application I proposed for this project is called the Smart Device Charging Hub Manager. It is based on the Smart Docklands the theme is smart mobility and a smart design/layout. In busy areas like Dublin Docklands many people use devices such as phones, laptops and electric scooters during the day for work, communication and travel. These devices need to be charged regularly and if they run out of battery it can cause problems for users who depend on them. One possible solution in a modern city is the use of public charging hubs where people can charge their devices while they are in the area. However, if there are many devices and limited charging ports it becomes important to manage the hubs properly so that devices can be charged efficiently and users do not have to wait too long. The aim of this application is to create a simple system that can manage charging hubs, devices waiting to charge and charging sessions. The system will allow a user to add and manage charging hubs by storing information such as a hub ID, location and number of charging ports available. Devices such as phones, laptops and scooters can also be registered in the system when they need to be charged. If all charging ports at a hub are full the device can be placed into a waiting list until a port becomes available. When a port becomes free the next device in the waiting list can begin charging. The application will also keep a record of charging sessions so that recent activity can be viewed. The program will be developed in Java using a Swing graphical user interface created in NetBeans so that users can interact with the system through simple forms and buttons. The user will be able to add charging hubs, add devices, start charging sessions and view stored data through the interface. The project will also show several programming concepts learned during the module. It will use inheritance and polymorphism by creating a base class called Device and subclasses such as Phone, Scooter and Laptop which all represent different types of devices that can be charged. An interface will also be used to define charging behaviour such as starting and stopping a charging session. The application will also make use of abstract data types. An Array List will store charging hubs and allow operations such as adding, viewing, updating and removing hubs. A Queue will manage devices waiting to charge so that they are processed in the order they arrive. A Stack will store recent charging sessions so that the most recent session can be accessed first. This application is a simple example of how software can be used to help manage public infrastructure in a smart city environment such as Dublin Docklands.

Class Diagram



https://github.com/Desmond1up/smart-docklands-charging-hub-manager